DATA, PLOTS, MATRICES

DATA MATRIX / STANDARDIZATION

$$X \in \mathbb{R}^{N\times M}, \qquad \tilde{x}_{ij} = \frac{x_{ij}-\bar{x}_j}{s_j}, \qquad \bar{x}_j = \frac{1}{N}\sum_i x_{ij}, \quad s_j^2 = \frac{1}{N-1}\sum_i (x_{ij}-\bar{x}_j)^2 \tag{F1}$$

MEANS: N = rows/observations; M = attributes; x_{ij} = value in row i, attribute j; $\bar{x}_j$ = column mean; s_j = sample standard deviation; $\tilde{x}_{ij}$ = standardized value. READ: $\tilde{x}_{ij} = 2$ means row i is 2 sd above average on attribute j. USE BEFORE: distances, PCA, ridge/logistic models when units differ.

COVARIANCE / CORRELATION / DISTANCE

$$\hat{\Sigma} = \frac{1}{N-1}\tilde{X}^\top \tilde{X}, \qquad \rho_{jk} = \frac{\mathrm{Cov}(x_j,x_k)}{s_j s_k}, \qquad d_2(x,z) = \sqrt{\sum_m (x_m - z_m)^2}, \quad d_1(x,z) = \sum_m |x_m - z_m|, \qquad d_\infty(x,z) = \max_m |x_m - z_m| \tag{F2}$$

MEANS: $\hat{\Sigma}$ = covariance matrix; for standardized data it is the correlation matrix; ρ_{jk} = correlation between attributes j,k; x,z = two rows/vectors; m indexes coordinates. READ MATRIX: diagonal covariance/correlation = variance/1; positive sign tilts scatter upward; negative sign tilts downward; $|\rho| \approx 0$ gives round/no linear trend; $|\rho|$ near 1 gives narrow line. READ DISTANCE MATRIX: smallest off-diagonal entry = nearest other row.

PLOT AND TABLE READING

OBJECT	HOW TO READ IT
HISTOGRAM	x-axis = value; high bars = many rows; tails/skew show extremes.
BOXPLOT	middle line = median; box = middle 50%; long whisker = skew/spread.
SCATTER	read axes first; upward/downward cloud gives correlation sign.
SCATTER MATRIX	panel (j,k) compares attributes j,k; match sign/strength to ρ_{jk}. Marginal shape is read from repeated axis panels/densities.
COVARIANCE ELLIPSE	diagonal entries set spread on axes; off-diagonal sign sets tilt; large weight in GMM means more points from that component.
TABLE OF COUNTS	denominator is the condition after the bar: for $P(A\|B)$ divide rows with both A,B by rows with B.

REGRESSION ERROR

$$e_i = y_i - \hat{y}_i, \qquad \mathrm{SSE} = \sum_i e_i^2, \qquad \mathrm{MSE} = \frac{1}{N}\sum_i e_i^2, \quad \mathrm{RMSE} = \sqrt{\mathrm{MSE}} \tag{F3}$$

MEANS: y_i = true numeric target; $\hat{y}_i$ = prediction; e_i = residual/error; N = number of tested rows. CALC: subtract prediction from truth, square, average. READ: smaller test/CV MSE or RMSE means better numeric prediction on unseen rows.

PCA / SVD / VECTORS

PCA OBJECTS

$$\tilde{X} = USV^\top, \qquad V = [v_1,\dots,v_M], \qquad Z = \tilde{X}V = US, \quad z_{ik} = \tilde{x}_i^\top v_k, \qquad \hat{\tilde{x}}_i = z_i V_K^\top \tag{F4}$$

MEANS: $\tilde{X}$ = centered/standardized data; U,S,V = SVD matrices; S has singular values s_k; v_k = column k of V, the PC k loading vector; Z = score matrix; z_{ik} = row i's coordinate on PC k; V_K = first K loading columns. READ V: read down column k; large $|v_{mk}|$ means attribute m defines PC k. Sign says direction only; all signs in one PC may flip with same meaning.

PCA PROJECTION / VARIANCE

$$z_k = \tilde{X}v_k, \qquad \mathrm{Var}(z_k) = \frac{1}{N-1}\sum_i z_{ik}^2 = \frac{v_k^\top \tilde{X}^\top \tilde{X} v_k}{N-1} = \frac{s_k^2}{N-1} \tag{F5}$$

MEANS: z_k = all scores on PC k; z_{ik} = score for row i; $N-1$ gives unbiased sample variance; s_k = singular value. CALC: center/standardize row, multiply matching entries by loading column, add. READ SCORE: positive score means row points in positive loading direction; negative score means opposite profile.

EXPLAINED VARIANCE

$$\lambda_k = \frac{s_k^2}{N-1}, \qquad \mathrm{EVR}_k = \frac{s_k^2}{\sum_j s_j^2}, \qquad \mathrm{cumEVR}(K) = \frac{\sum_{k=1}^K s_k^2}{\sum_j s_j^2} \tag{F6}$$

MEANS: λ_k = variance on PC k; EVR = fraction of total variance explained by one PC; cumEVR = fraction explained by first K PCs. CALC: square singular values, divide by total, accumulate. USE: choose smallest K reaching required variance.

PCA PLOTS AND OUTPUT

OBJECT	READ IT
LOADING COLUMN	variable weights for one PC; largest absolute entries name what the PC measures.
SCORE PLOT	each point is one row in PC coordinates; nearby points have similar standardized profiles.
LOADING PLOT	variable arrows/points; same direction = similar contribution; opposite = negative relation.
BIPLOT	project an observation along a variable arrow; far in arrow direction = high value of that variable.
PC DIRECTIONS	columns of V are unit length and mutually perpendicular: $v_a^\top v_b = 0$, $\|v_k\| = 1$.

NORM / DOT PRODUCT / COSINE

F7
$$\|x\|_2 = \sqrt{x^\top x}, \qquad x^\top y = \sum_m x_m y_m,$$
$$\cos(x,y) = \frac{x^\top y}{\|x\|_2\|y\|_2}, \quad \|X\|_F^2 = \sum_{ij} x_{ij}^2 = \sum_k s_k^2$$

MEANS: $\|x\|_2$ = vector length; $x^\top y$ = dot product; cosine = direction similarity; $\|X\|_F$ = matrix length. READ: cosine near 1 = same direction; near 0 = perpendicular; near -1 = opposite direction.

PROBABILITY / BAYES / DENSITY

BAYES RULE

F8
$$P(A|B) = \frac{P(A,B)}{P(B)},$$
$$P(y=c|x) = \frac{P(x|y=c)P(y=c)}{\sum_{c'} P(x|y=c')P(y=c')}$$

MEANS: A, B = events; x = observed attributes; y = class label; c, c' = possible classes; $P(y=c)$ = prior/class frequency before seeing x; $P(x|y=c)$ = likelihood/density of x inside class c; $P(y=c|x)$ = posterior after seeing x. CALC: for each class, score = likelihood × prior; divide each score by sum of all class scores. READ: equal likelihoods do not imply equal posteriors if priors differ.

COUNT TABLE / NAIVE BAYES

F9
$$\hat{P}(y=c|x=a) = \frac{\#\{i : y_i = c, x_i = a\}}{\#\{i : x_i = a\}},$$
$$\text{score}(c) = P(y=c) \prod_{m=1}^{M} P(x_m|y=c),$$
$$P(c|x) = \frac{\text{score}(c)}{\sum_{c'} \text{score}(c')}$$

MEANS: $\#\{\}$ = count rows satisfying condition; a = specific value; x_m = attribute m of row x; score = unnormalized posterior. READ TABLE: for $P(x_m = a|y=c)$ divide by number of rows in class c. NAIVE means multiply feature probabilities as if conditionally independent given the class.

GAUSSIAN DENSITY CLASSIFIER

F10
$$\mathcal{N}(x|\mu,\Sigma) = \frac{\exp[-\frac{1}{2}(x-\mu)^\top \Sigma^{-1}(x-\mu)]}{(2\pi)^{M/2}|\Sigma|^{1/2}},$$
$$\mathcal{N}(x|\mu,\sigma^2) = \frac{e^{-(x-\mu)^2/(2\sigma^2)}}{\sqrt{2\pi}\sigma},$$
$$\hat{y} = \arg\max_c \mathcal{N}(x|\mu_c,\Sigma_c)P(c)$$

MEANS: μ = mean/center; Σ = covariance matrix; σ = 1D standard deviation; $|\Sigma|$ = determinant; Σ^{-1} weights distance by covariance. READ CURVE: highest near μ; larger variance = wider and lower. READ SCATTER: covariance sign gives tilt; diagonal entries give spread of coordinates. DENSITY HEIGHT is not a probability alone; compare density × prior.

GMM / EM RESPONSIBILITY

F11
$$p(x) = \sum_{k=1}^{K} \pi_k \mathcal{N}(x|\mu_k,\Sigma_k),$$
$$\gamma_{ik} = P(z_i = k|x_i) = \frac{\pi_k \mathcal{N}(x_i|\mu_k,\Sigma_k)}{\sum_j \pi_j \mathcal{N}(x_i|\mu_j,\Sigma_j)}$$

MEANS: K = number of components; π_k = mixing weight/prior of component k; z_i = hidden component label; γ_{ik} = responsibility/posterior for row i and component k. E-STEP: compute all γ_{ik} using current parameters. READ: responsibilities for one row sum to 1; largest responsibility gives hard component assignment; large π_k means many expected points from component k.

KDE / LEAVE-ONE-OUT DENSITY

F12
$$\hat{p}(x) = \frac{1}{N}\sum_{j=1}^{N} \mathcal{N}(x|x_j,\sigma^2 I),$$
$$\hat{p}_{\text{LOO}}(x_i) = \frac{1}{N-1}\sum_{j\neq i} \mathcal{N}(x_i|x_j,\sigma^2 I)$$

MEANS: KDE = average of Gaussian kernels centered at training rows; σ = kernel width; I = identity covariance. CALC: for LOO, use all kernel values except the row's own kernel. READ: larger σ smooths more; small density means unusual compared with training rows.

PREDICTION / BOUNDARIES

LINEAR / RIDGE REGRESSION

F13
$$\bar{x} = [1, x_1, \ldots, x_M]^\top, \qquad \hat{y} = \bar{x}^\top w,$$
$$w_{\text{LS}} = (X_a^\top X_a)^{-1} X_a^\top y,$$
$$w_\lambda = (X_a^\top X_a + \lambda D)^{-1} X_a^\top y$$

F14
$$E_\lambda(w) = \sum_i (y_i - \bar{x}_i^\top w)^2 + \lambda \sum_{j=1}^{M} w_j^2,$$
$$D = \text{diag}(0, 1, \ldots, 1)$$

MEANS: $\bar{x}$ = input with leading 1 for intercept; X_a = data with intercept column; w_0 = intercept; w_j = slope/weight; λ = regularization strength; D usually leaves intercept unpenalized. CALC: standardize test point if training used standardized data, insert leading 1, multiply by weights, add. READ: positive w_j increases prediction when x_j increases. REGULARIZATION: larger λ shrinks weights, raises bias, usually lowers variance; more features/polynomials lower bias, often raise variance.

LOGISTIC REGRESSION / PROBABILITY

$$a = \bar{x}^\top w,$$
$$\hat{p} = P(y=1|x) = \sigma(a) = \frac{1}{1+e^{-a}}, \quad \text{(F15)}$$
$$\hat{y} = \mathbb{I}[\hat{p} \geq \tau], \quad \log\frac{\hat{p}}{1-\hat{p}} = a$$

MEANS: a = score/log-odds; $\hat{p}$ = predicted probability of stated positive class; τ = threshold; σ = sigmoid. READ: at $\tau = 0.5$, boundary is $a = 0$; positive a gives $\hat{p} > 0.5$. Positive weight raises positive-class probability, holding other inputs fixed.

FEATURE MAP / HEATMAP / BOUNDARY

$$\tilde{x} = \phi(x),$$
$$\hat{y} = w^\top \tilde{x} \quad \text{(regression)}, \quad \text{(F16)}$$
$$P(y=1|x) = \sigma(w^\top \tilde{x}) \quad \text{(classification)}$$

MEANS: $\phi(x)$ = transformed features, e.g. $[1, x_1, x_2^2]$; $\tilde{x}$ = feature vector after transformation. READ HEATMAP: contours follow constant $w^\top \tilde{x}$; linear features give straight contours; squared/cubic/sine/cos features give curved/repeated contours; huge regularization gives nearly flat output.

K-NEAREST NEIGHBOURS

$$\hat{y}_{\text{reg}}(x) = \frac{1}{K}\sum_{i \in N_K(x)} y_i, \quad \text{(F17)}$$
$$\hat{y}_{\text{class}}(x) = \arg\max_c \sum_{i \in N_K(x)} \mathbb{I}[y_i = c]$$

MEANS: $N_K(x) = K$ training rows closest to test row x using chosen distance; c = class. CALC: compute distances, sort, keep K, average targets or count class votes. READ BOUNDARY: small K = local/jagged; large K = smoother. If tied classes, use the exam's stated tie rule; otherwise nearest among tied classes is common.

VALIDATION / METRICS / MODEL OUTPUT

HOLD-OUT / K-FOLD / NESTED CV

$$E_T = \frac{1}{|T|}\sum_{i \in T} L(y_i, \hat{f}(x_i)), \quad E_{\text{CV}} = \frac{1}{K}\sum_{k=1}^{K} E_k, \quad \text{(F18)}$$
$$\hat{E}_{\text{gen}} = \frac{1}{K_1}\sum_{i=1}^{K_1} E^{\text{test}}_{s_i^*, i}$$

MEANS: T = test/held-out set; L = loss; E_k = fold validation error; K_1 = outer folds; s_i^* = model/complexity selected by inner CV inside outer fold i; $E^{\text{test}}_{s_i^*, i}$ = outer test error of that selected model. CALC NESTED: inner folds choose model in each outer fold; outer test errors estimate generalization; average only the chosen model's outer errors.

CONFUSION MATRIX

F19

	$\hat{y} = +$	$\hat{y} = -$
$y = +$	TP	FN
$y = -$	FP	TN

MEANS: $+$ = stated positive class; TP = actual positive predicted positive; FN = actual positive predicted negative; FP = actual negative predicted positive; TN = actual negative predicted negative. READ: do not trust row/column position blindly; identify actual class and predicted class from table labels.

CLASSIFICATION METRICS

$$\text{Acc} = \frac{TP+TN}{N}, \quad \text{Err} = 1 - \text{Acc},$$
$$\text{Recall} = \text{TPR} = \frac{TP}{TP+FN}, \quad \text{FPR} = \frac{FP}{FP+TN}, \quad \text{(F20)}$$
$$\text{Precision} = \frac{TP}{TP+FP}, \quad F_1 = \frac{2PR}{P+R}$$

MEANS: $N = TP + FN + FP + TN$; P = precision; R = recall. READ: recall = fraction of actual positives found; precision = fraction of predicted positives correct; FPR = fraction of actual negatives wrongly predicted positive.

ROC / AUC / THRESHOLD

$$\text{ROC}(\tau) = (\text{FPR}(\tau), \text{TPR}(\tau)), \quad \text{(F21)}$$
$$\text{AUC} = \text{area under ROC}$$

MEANS: threshold τ turns scores/probabilities into classes; ROC x-axis = FPR; ROC y-axis = TPR. CALC FROM SORTED SCORES: start at $(0, 0)$ with no predicted positives; lowering threshold over a true positive moves up; over a true negative moves right. READ: upper-left is best; random ranking has AUC near 0.5; high threshold usually predicts fewer positives.

TREES / ADABOOST / ANN

DECISION TREE / HUNT SPLIT

$$I_{\text{error}} = 1 - \max_c p_c,$$
$$I_{\text{Gini}} = 1 - \sum_c p_c^2,$$
$$I_{\text{ent}} = -\sum_c p_c \log_2 p_c, \quad \text{(F22)}$$
$$\text{gain} = I(\text{parent}) - \sum_v \frac{N_v}{N} I(\text{child } v)$$

MEANS: $p_c = n_c/N_{\text{node}}$ = class fraction in node; I = impurity; v = child branch; N_v/N = child size weight. CALC: count labels in parent and children, compute impurity, subtract weighted child impurity, choose largest gain. READ TREE: follow inequalities root to leaf; a leaf region is intersection of all split rules on that path; classification leaf predicts majority/class fractions.

ADABOOST: CLASS-LABEL VERSION

$$\epsilon_t = \sum_i q_i^{(t)} \mathbb{I}[h_t(x_i) \neq y_i], \quad \alpha_t = \frac{1}{2}\log\frac{1-\epsilon_t}{\epsilon_t}, \quad \text{(F23)}$$
$$\tilde{q}_i^{(t+1)} = \begin{cases} q_i^{(t)} e^{-\alpha_t}, & h_t(x_i) = y_i \\ q_i^{(t)} e^{\alpha_t}, & h_t(x_i) \neq y_i \end{cases}$$

$$\text{F24} \quad H(x) = \arg\max_c \sum_t \alpha_t \mathbb{I}[h_t(x) = c]$$

MEANS: $q_i^{(t)}$ = row weight before round t; ϵ_t = weighted error; α_t = classifier vote weight; $\tilde{q}$ = unnormalized updated weight; normalize only if next round needs weights summing to 1. READ: wrong rows get larger weight; correct rows get smaller weight when $\epsilon_t < 0.5$. More rounds can overfit if weak learners are flexible.

ANN FORWARD PASS

$$\text{F25} \quad \begin{aligned} a_j &= w_{0j}^{(1)} + \sum_m w_{mj}^{(1)} x_m, \qquad h_j = g(a_j), \\ \hat{y} &= w_0^{(2)} + \sum_j w_j^{(2)} h_j, \quad \mathrm{ReLU}(a) = \max(0, a), \\ \sigma(a) &= \frac{1}{1 + e^{-a}} \end{aligned}$$

MEANS: a_j = input to hidden unit j; $w_{0j}^{(1)}$ = hidden bias; $w_{mj}^{(1)}$ = input-to-hidden weight; h_j = hidden activation; g = activation; $w^{(2)}$ = hidden-to-output weights. CALC: compute all hidden a_j, apply activation, multiply hidden outputs by output weights, add output bias. OUTPUT: linear for real-valued regression; sigmoid for binary probability; softmax for multiclass; ReLU only for nonnegative target.

ANN PARAMETER COUNT

$$\text{F26} \quad \#w = (M+1)n_h + (n_h+1)C$$

MEANS: M = input attributes; n_h = hidden units; C = output units/classes; '+1' counts bias. For one output, $\#w = (M+1)n_h + (n_h+1)$. READ WEIGHTS: large positive input weight increases hidden activation; large negative weight decreases it; bias is added even when inputs are zero.

CLUSTERING / ASSOCIATION

K-MEANS

$$\text{F27} \quad \min_{\{C_k, \mu_k\}} \sum_{k=1}^{K} \sum_{i \in C_k} \|x_i - \mu_k\|^2, \quad \mu_k = \frac{1}{|C_k|} \sum_{i \in C_k} x_i$$

MEANS: C_k = rows assigned to cluster k; μ_k = centroid/mean vector; $|C_k|$ = cluster size. CALC: assign each row to nearest centroid, recompute centroid as coordinate-wise mean, repeat if asked. READ CENTROID TABLE: high standardized coordinate = cluster is above average on that attribute; cluster labels have no natural order.

HIERARCHICAL CLUSTERING

LINKAGE	CLUSTER DISTANCE
SINGLE / MINIMUM	smallest pair distance across two clusters; can chain.
COMPLETE	largest pair distance; compact clusters.
AVERAGE	average of all cross-cluster pair distances.
CENTROID	distance between cluster means.

READ DENDROGRAM: leaves = rows; merge height = distance where groups join; first merge has smallest distance. CALC: to get K clusters, cut horizontally so K branches remain.

RAND / JACCARD CLUSTER AGREEMENT

$$\text{F28} \quad \mathrm{Rand} = \frac{f_{11} + f_{00}}{\binom{N}{2}}, \quad J = \frac{f_{11}}{f_{11} + f_{10} + f_{01}}$$

MEANS: inspect row pairs; f_{11} = together in both clusterings; f_{00} = apart in both; f_{10}, f_{01} = together in one but apart in other. READ: Rand counts both same-cluster and different-cluster agreement; Jaccard ignores shared separation.

LOCAL DENSITY / OUTLIER

$$\text{F29} \quad \begin{aligned} \mathrm{dens}(x_i, K) &= \left(\frac{1}{K} \sum_{x_j \in N_i} d(x_i, x_j) \right)^{-1}, \\ \mathrm{ARD}(x_i, K) &= \frac{1}{K} \sum_{x_j \in N_i} \frac{\mathrm{dens}(x_i, K)}{\mathrm{dens}(x_j, K)} \end{aligned}$$

MEANS: N_i = K nearest neighbours of row i; density = inverse average neighbour distance; ARD = density relative to neighbours. READ: small density or small ARD is outlying; ARD is useful when clusters have different density.

ASSOCIATION RULES / APRIORI

$$\text{F30} \quad \begin{aligned} \mathrm{supp}(X) &= \frac{\#(X)}{N}, \\ \mathrm{conf}(X \to Y) &= \frac{\mathrm{supp}(X \cup Y)}{\mathrm{supp}(X)}, \\ \mathrm{lift}(X \to Y) &= \frac{\mathrm{conf}(X \to Y)}{\mathrm{supp}(Y)} \end{aligned}$$

MEANS: X, Y = disjoint itemsets; X = left side/antecedent; Y = right side/consequent; $\#(X)$ = baskets containing all items in X; N = baskets. CALC: count baskets with X, Y, and $X \cup Y$. READ: support = frequency; confidence = $P(Y|X)$; lift > 1 means Y is more common after seeing X than baseline. APRIORI: if an itemset is below minimum support, all supersets are also below.